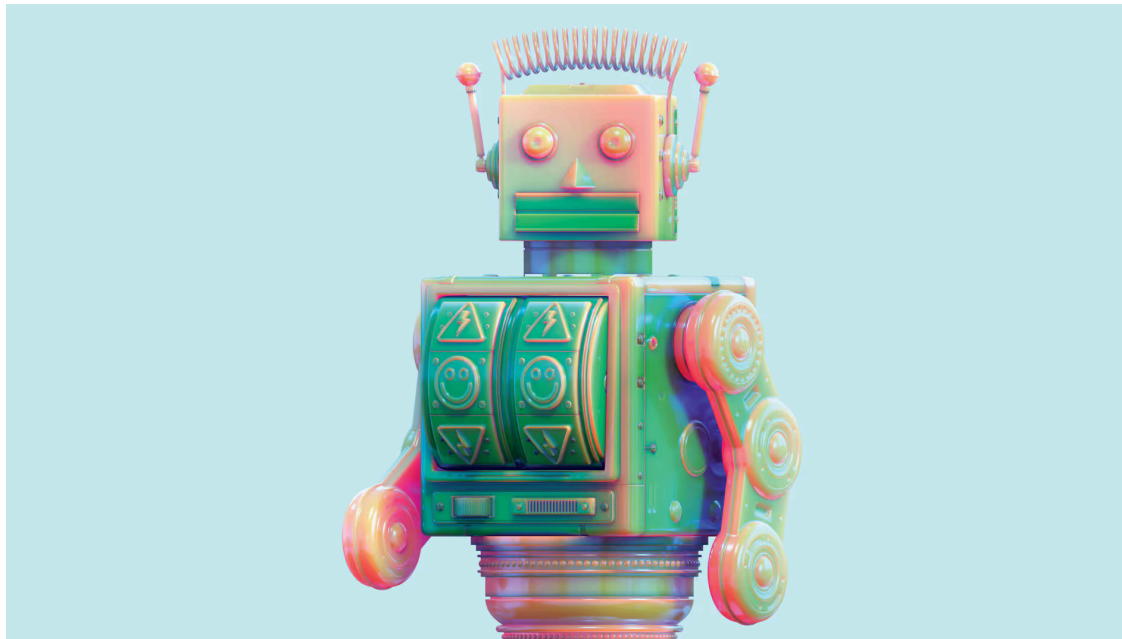


Generative AI

# How Gen AI Robots Are Reshaping Services

by Jochen Wirtz

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**Summary.** By incorporating generative AI, robots are evolving from scripted machines to adaptive systems that interpret context, learn from demonstrations,



and adjust their behaviors in real time. Advances in large language models and related technologies are... [more](#)

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**If you've had the chance** to ride in a Waymo, you've likely emerged from the vehicle amazed by its abilities. Since Alphabet launched the project, in 2009, Waymo has developed a fleet of 2,500 driverless robotaxis now on the road in San Francisco, Miami, Phoenix, and other cities, where they've completed more than 20 million trips. The vehicles do more than zip passengers around at up to 65 miles an hour. They can respond to verbal instructions ("Play some '80s rap on Spotify") or answer questions ("What time is the Giants game?") while switching lanes to avoid a double-parked delivery van. Early customer service data and comments on the Waymo One app show that riders are thrilled by the experience.



Waymo is a specific use case of a technology that's maturing rapidly and set for significant deployment: robots powered by generative AI. Many companies are already using gen AI chatbots, agents, and related technologies to automate and scale up customer service, but in most of these cases customers interact with the technology on screens. Embedding gen AI into robots gives companies the chance to reinvent their interactions with customers in physical settings—restaurants, hotels, hospitals, retail stores, and other brick-and-mortar locations—where service has remained stubbornly human. Using large language models (LLMs), large behavioral models (LBMs), and agentic AI, this new generation of robots can better understand context, make inferences, and provide personalized experiences. They can converse like competent employees—following logic across conversational turns, clarifying ambiguity, and explaining complex ideas simply. In the past a patient asking a robot, “Is this going to hurt? How long will it take? And what happens if I feel dizzy?” would overwhelm the limits of its script. But today's LLM-powered service robots can unpack those concerns and offer plain-language responses. A robot named Robin is already doing just that, providing emotional support in 30 pediatric units and nursing homes across the country. It moves around autonomously to greet children and answer questions. Nurses can give Robin verbal commands, such as “Go to room 517 for 20



minutes, and then go to room 516 for 10 minutes.” It also comes loaded with games children can play using spoken responses.

This probably isn't the first article you've read claiming that robots will transform the service sector. Admittedly, their impact has grown more slowly than enthusiasts expected. The global market for professional service robots, which includes models for logistics, healthcare, cleaning, and other sectors, grew roughly 9% in 2024, reaching almost 200,000 units sold. But many pilot projects stalled or underperformed. McKinsey research shows that 71% of companies say that high up-front costs are a major challenge when adopting robots, and 61% point to a lack of experience with automation as another top hurdle. Maintenance and reliability remain ongoing challenges, as does customer and employee acceptance; many people still prefer human interaction, especially in situations that are complex or emotionally charged. When companies have deployed customer-facing robots, most implementations have been confined to narrowly scripted tasks such as delivering room service or baggage in a hotel. These service robots are not much better than elaborate mobile vending machines. They dependably follow preprogrammed routes, read barcodes, and answer FAQs but largely have failed to deliver the scale or returns that early adopters had hoped for.



Nonetheless, virtually every major robot manufacturer is integrating gen AI into its offerings—and some of the early results show real promise. I’ve been researching advanced robotics for more than a decade, and over the past 18 months I’ve traveled to Europe, Asia, and North America to observe deployments of physical service robots powered by gen AI in 14 organizations operating in financial services, healthcare, education, hospitality, and more. In this article I’ll outline how firms can use this new technology to create value, mitigate risks, and build the organizational muscle for its success.

## **What Are Gen AI Robots?**

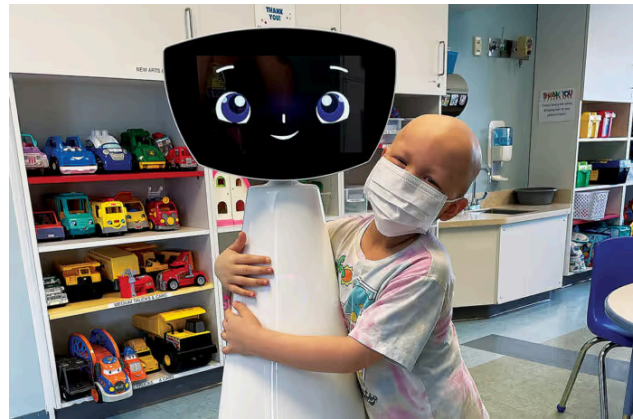
Gen-AI-enabled robots rely on a mix of technologies—some that are familiar and some that are not.

By now most executives have a grasp of LLMs and agentic AI. In a robot, LLMs allow for conversation, and agentic AI adds memory, planning, execution, and reflection. Using those technologies, a robot can remember a returning customer, reason through trade-offs (*Is early check-in possible if housekeeping says the room is almost ready?*), plan a sequence of tasks, execute steps across digital systems and physical spaces, and then reflect on what worked and what didn’t. The difference between traditional, script-bound robots and those powered by agentic AI is night and day. A script-bound system might recognize that a delivery must



be prioritized but still follow a predetermined route. An agentic system can assign the task to a human, reroute the delivery, and reallocate resources to enable it. While a robot has domain-specific intelligence (one in a hotel will be trained very differently from one in a hospital), it can make complex decisions to execute high-level instructions, such as “check in guests quickly” or “replenish the inventory of IVs by the end of the shift.”

## AI Robots Adding Value Across Industries



Robin, a therapeutic robot, comforts a pediatric patient at UMass Memorial Medical Center. Courtesy of Robin the Robot

LBM is a less familiar technology. They are trained on large sets of behaviors, just as LLMs are trained on a seemingly infinite supply of text. LBMs help robots deal with the fact that service in a physical environment is messy. Trays tilt. Floors get slick. Customers hand over fragile items. As such, programming robots for every contingency is impossible. Instead, developers teach robots to learn using LBMs for any contexts that are required. LBMs are what allow Waymo vehicles to drive around double-parked vans.

Gen AI robots use cameras, microphones, and sensors to learn by observing humans, by asking questions, and through trial and error. This training can initially be done in the real world. Robots can acquire behaviors by observing a handful of demonstrations (of how to carefully pick up a glass of wine, for example) and then experimenting with millions of microvariations of speed, grip, and trajectory, using the metaverse or a digital twin to perfect their approach. The behaviors can also be transferred across contexts. If a robot learns to handle a fragile glass in a café, elements of that skill will carry over to handling vials in a clinic or delicate merchandise in a boutique.

Other technologies facilitate gen AI robots' ability to learn. No-code programming and fleet learning (sharing instruction across a group of robots), for instance, make implementation and



improvements easier than they were a decade ago. In the past any adjustment to robots required a ticket to IT or a visit from a vendor. No-code training (made possible by gen AI) lets frontline workers adjust a robot's behavior by asking why it took a certain approach, describing a better one, and physically demonstrating it. This ease of improvement compresses the cycle time for operational enhancements from months to days. These capabilities turn robots into adaptive systems that integrate conversation, cognition, and physical action—at scale and personalized for each customer.

Consider the robots at work inside BMW's automobile assembly plant in Spartanburg, South Carolina. The auto industry was an early adopter of robotics; factories have used simple robotic arms to do repetitive tasks (such as spot welding) since the 1980s. But in 2024 BMW began piloting [Figure 02](#), a humanoid robot that represents a sharp break from traditional industrial automation. Unlike conventional factory robots, Figure 02 can move autonomously through the plant using six onboard cameras, interpret what it sees, and reason about how objects should be used, drawing on a large base of automotive and general knowledge. Powered by OpenAI models, it listens to and processes human speech, infers intent from even vague instructions, asks clarifying questions when needed, and learns from its mistakes over time. During an 11-month deployment,





Figure 02 contributed to the production of roughly 30,000 BMW vehicles. It acted as a high-precision pair of hands in the body shop, carrying and placing fragile sheet-metal parts and lining components up so that the welding robots could build car frames. BMW is now upgrading to Figure 03, a lighter, taller successor designed to apply these capabilities beyond the factory floor. In a [promotional video](#), Figure 03 performs tasks such as washing dishes, folding laundry, serving drinks, and playing fetch—highlighting that gen AI is giving robots the ability to take on an expanding range of functions.

## How to Deploy Gen AI Robots

My work with companies shows that the most exciting uses of gen AI robots involve frontline and customer- and employee-facing tasks. Bringing a robot into a workplace is more complicated than just unboxing a gadget, because many workplaces are unpredictable—waiters carry trays, doctors and nurses hustle from room to room, and so on. To convert potential into performance, leaders must carefully select use cases, communicate to customers and employees why and how they're using robots, and set up guardrails. Four critical steps, based on my observations in the field, can guide them.

**1. Start with use cases that address labor constraints.** Robots are most effective when applied to repeatable, economically valuable



tasks that provide measurable returns. Many of the early experiments I observed involved roles in industries that face chronic labor shortages. That makes sense: The robots are reducing not only costs but also the difficulty of recruiting hard-to-find workers. Once you've identified target roles, start by examining the specific tasks those jobs require. Are they repeatable enough for a robot to learn quickly? Is the payoff from turning them over to a robot immediate in terms of speed, efficiency, consistency, or freeing up employees to do work where they add more value?

Good candidates for experiments include check-in and checkout in hotels, order modification and delivery in quick-service restaurants, and logistics in hospital wards—settings where there's constant pressure on frontline capacity and a clear operational handoff the robot can own.

Once you've chosen the use case, design the pilot so that frontline employees can actively improve performance in the flow of work. An employee who notices a robot carrying out a suboptimal sequence should be able to explain and demonstrate a better method—without waiting weeks for a software update.

Done well, this approach expands what robots can do over time, enhances their performance in the domain, and makes their



implementation more effective—while shifting frontline roles toward more-skilled work and easing labor shortages where they're most acute.

**2. Design robot interactions for customer acceptance.** Most resistance to robots begins at the customer touchpoint, not with the technology itself. Traditional kiosks, self-service technologies, and scripted bots often force customers into rigid, unnatural sequences. This is a design problem, and LLM-enabled robots undo that burden. Customers and employees can speak naturally to them and—critically—the robots can follow through in the physical environment.

Consider hotel check-in. A human employee typically navigates multiple systems—reservation records, loyalty profiles, housekeeping status, payment, and preferences—one screen at a time. A gen-AI-powered robot can connect to those systems in parallel, reconcile conflicts, and deliver a room key in seconds while interacting with and welcoming the guest. To the customer, the experience is a warm, efficient conversation—and often proves more consistent than human service under pressure.

Early deployments highlight both the promise and the pitfalls. At Henn na Hotel in Tokyo, robot receptionists guide guests through identity verification, room assignment, and payment. The



initiative has helped with labor costs and shortages, but not everything has worked as planned. At times the robots have struggled with accents, background noise, and unexpected requests, sometimes increasing rather than reducing staff workload.

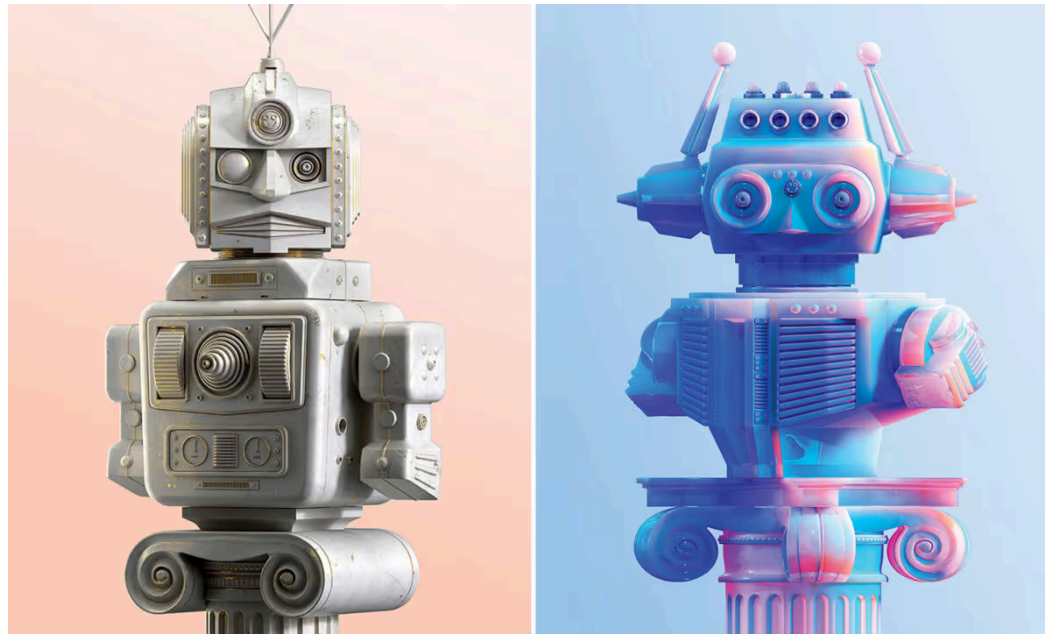
Designing for customer acceptance means testing these interactions in real environments with real customers, learning where friction arises, and ensuring the robot's conversational competence is matched by reliable physical execution.

**3. Position robots as service enhancers, not workforce replacements.** How robots are introduced—and how their role is explained—strongly shapes how people perceive them. Acceptance of AI varies by demographics and context, and today many customers still prefer human interaction for encounters that require warmth, empathy, or judgment. Employees, meanwhile, are worried about losing their jobs to the technology.

To help address those preferences and fears, companies should position gen AI robots not as replacements for frontline employees but as tools that improve accessibility, speed, and reliability while freeing humans to focus on emotionally charged, ambiguous, or high-stakes interactions. In my observations the most successful deployments happen where convenience



dominates customers' priorities and the value of automation is obvious.



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Companies should also clearly communicate what the robot does and how customers can access human support when needed. In hospitals, for example, robots could explain that they make simple interactions more comfortable and frictionless and that they call for a human when they reach their limits. In retail settings, visible floor associates paired with robotic backroom automation could help keep the shopping experience grounded in human presence.



Setting expectations honestly is critical. Share with customers and employees the things that robots excel at today—speed, consistency, and long-tail reliability—and where humans still outperform them, particularly in situations requiring empathy, creativity, or novel problem-solving. Use evidence rather than hype to reinforce the message. Metrics are more persuasive than abstract promises: “Our wait time dropped from six minutes to 90 seconds” or “Our nurses are walking 3,000 fewer steps per shift and spending an extra hour with patients.” Numbers like these demonstrate how robots constitute a win-win for customers and employees.



**4. Continually update responsible-use guidelines.** When companies roll out gen AI robots, the stakes around ethics, fairness, and privacy increase. Many robots come with cameras and microphones, which means they can digitize almost everything happening in face-to-face interactions (unless laws say otherwise). Add the capabilities of advanced language models, and suddenly robots can persuade people. That's useful for selling and upselling, but it can cross a line if customers are analyzed, nudged, and manipulated in ways the company never intended. Things get especially sensitive when robots make allocation or pricing decisions (such as upgrades, early check-ins, or personalized offers) that can create perceptions of bias or unfairness.

Successful deployments treat gen AI robots as learning systems, not finished products. Organizations need mechanisms to observe how people actually interact with robots, identify where interactions break down, and systematically improve performance over time. That begins with capturing agreed-upon signals—such as where users hesitate, repeat themselves, or override the robots—and turning those insights into clear, actionable takeaways for supervisors and frontline leaders.

There are also learning risks. Gen AI robots learn from interactions, and that creates opportunities for employees and



customers to take advantage of them. Without safeguards, robots can pick up behaviors you don't want. I've heard robots repeat inappropriate language in hotels, and I've seen museum visitors try to tamper with robots for their own entertainment. No-code tools and fleet learning add challenges. Employees can introduce errors or biased logic without meaning to, and malicious actors can sabotage robots—with bad behaviors spreading fast.

Improvement should be deliberate and iterative: Test changes, measure outcomes, keep what works, and repeat. In practice, no robot implementation works well at the outset, and careful fine-tuning is essential. Frontline employees play a central role in this learning loop and should be supported as citizen developers, and with appropriate microlearning processes, certifications, and no-code tools, domain experts can identify failure points and explain better approaches directly to robots, using natural language. Over time, supervisory roles shift from monitoring tasks to orchestrating systems—reviewing improvements, enforcing safety and consent protocols, and aligning incentives with quality, reliability, and customer outcomes beyond speed alone.

At the same time, learning must be bounded by strong governance and corporate digital responsibility. Because robots collect sensitive data, privacy is essential. And because they navigate physical environments—where they can collide with





people or property—safety is also a primary concern. Careful movement, environmental cleanliness, and reliability should be treated as nonnegotiables, backed by hazard analyses, geofencing (restricting a robot’s movement to a specific area), emergency stops (the ability for humans to halt the robot’s movement), and clear service level agreements with vendors for maintenance and incident response.

None of this means brands should avoid gen AI robots. But it does mean that safety, security, and responsible design need to be first-order priorities. Mistakes will happen. If your architecture, policies, and testing are strong, those mistakes are far less likely to turn into real harm. Good practices ensure that as robots become more capable, organizations remain accountable—scaling up innovation without sacrificing trust.

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Gen-AI-enabled robots provide a practical path toward achieving cost-effective service in the physical world. They can deliver consistency and personalization at scale, which has long been the Achilles’ heel of physical service. But gen AI robots require a complex and lengthy implementation—and because it must take place in real-world settings, the stakes are higher, failures are public, and physical safety becomes a major concern.



Although the technology is in its infancy, early pilots in the field show that gen-AI-powered robots can benefit businesses, employees, and customers alike. By following the steps outlined in this article, companies can begin deploying these robots in a structured and safe manner.

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**Jochen Wirtz** is a professor of marketing and the vice dean, MBA programs at the NUS Business School at the National University of Singapore.



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